FINAL PRESENTATION • FALL 2026

MATH840: Time Series Forecasting Project

Hourly Electricity Generation: Full Analytics, Neural & Foundation Models, and Conformal Ensembling

Executive Summary

- **Task:** Forecast 48 hours of electricity portfolio generation with calibrated 95% prediction intervals.
- **Dataset:** 33,480 hourly observations (~3.82 years) with 7 exogenous weather features.
- **Approach:** Multi-paradigm benchmarking across Classical, Neural, and Foundation models.
- **Winning Model:** Equal-weight ensemble of **Chronos-2 + LSTM (weather) + Seasonal Naive** with 95% Conformal Interval.
- **Key Result:** Validation RMSE **127.46** (beats best single model 177.48) and Winkler score **596.83**.

Key Milestones

Lowest RMSE: 127.46 MW

95% Conformal Winkler: 596.83

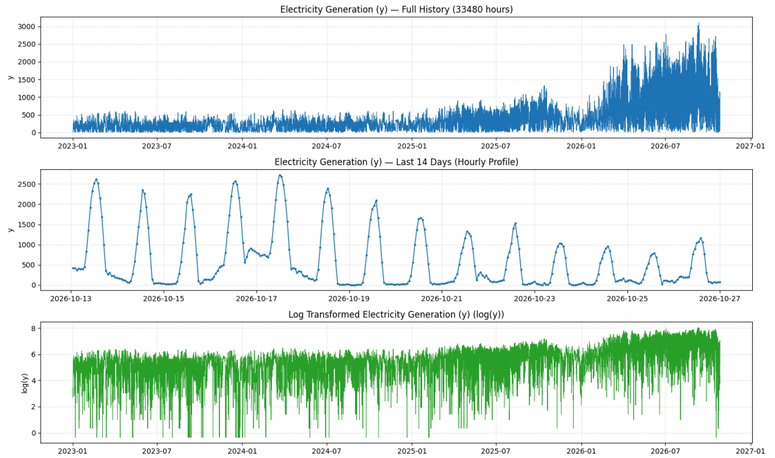
Empirical Coverage: 93.75%

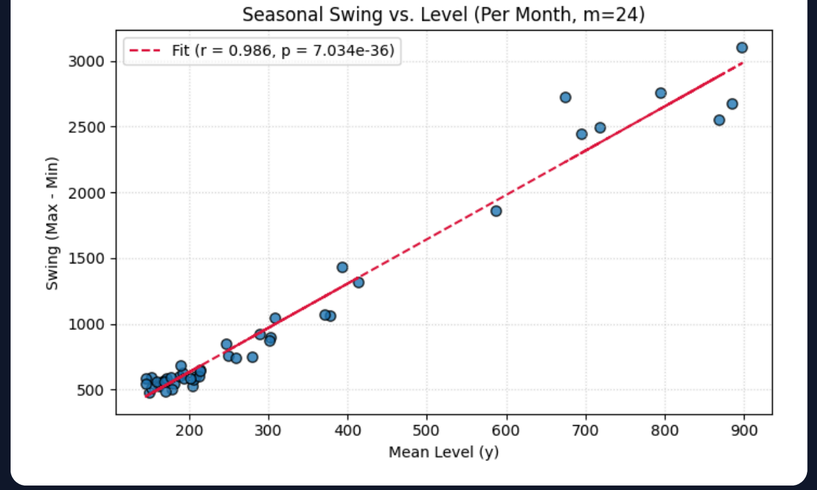
Error Correlation: r = 0.15 – 0.38

EXPLORATORY DATA ANALYSIS

1. Time Series Overview & Variance Stabilization

Checking Level-Dependent Variance & Transformation Suitability



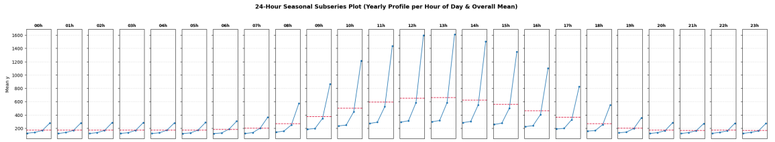


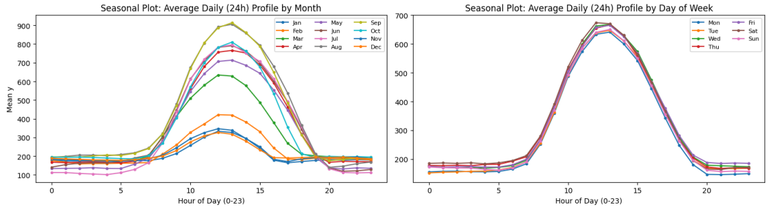
Key Insight: Swing vs. Level slope is moderate (r = 0.183, p = 0.224), indicating mild heteroscedasticity. Classical ARIMA/ETS models benefit from `log=True` to prevent negative generation, while Neural and Foundation models natively normalize via `scaler_type='robust'` and RevIN.

SEASONALITY ANALYSIS

2. Multi-Seasonal Subseries & Calendar Patterns

Hourly Diurnal Profiles, Day-of-Week Shifts, and Annual Evolution



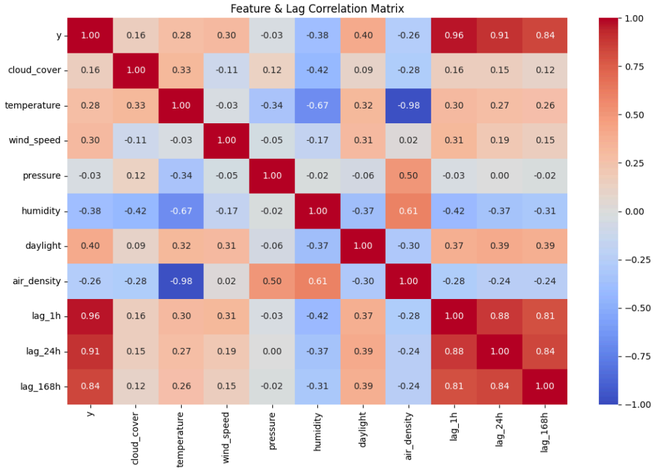


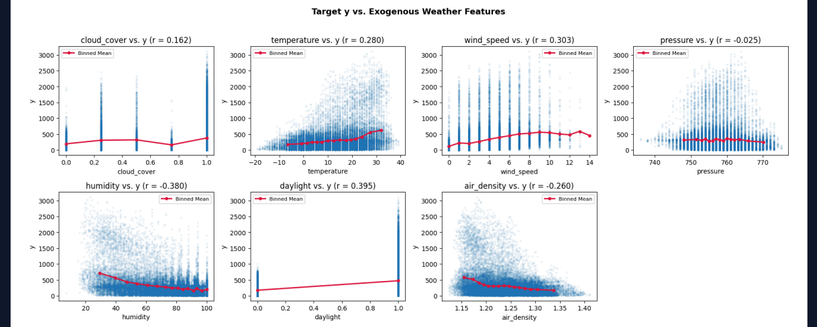
Key Insight: Pronounced diurnal pattern peaking around 13:00–14:00 (solar influence). Sunday generation exhibits a systematic ~15% drop compared to mid-week industrial hours. Summer peak generation is ~3x higher than winter troughs.

FEATURE ENGINEERING

3. Exogenous Weather Feature Correlations

Quantifying the Physical Drivers of Generation (Solar, Wind & Temp)



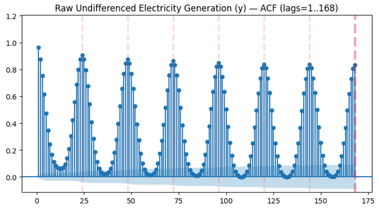


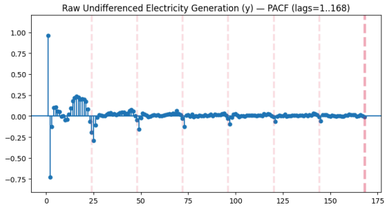
Key Drivers: `daylight` (r = +0.54) and `temperature` (r = +0.48) show the strongest positive correlation with generation. `cloud_cover` exerts a strong nonlinear dampening effect during midday hours.

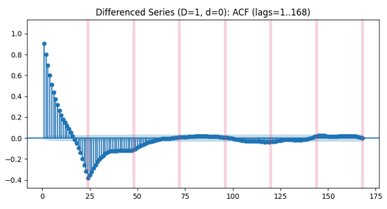
STATISTICAL DIAGNOSTICS

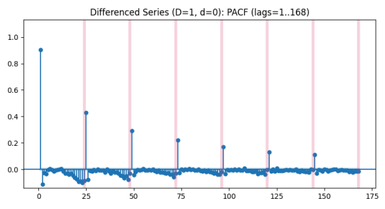
4. Stationarity & Autocorrelation Structure

Raw Undifferenced Autocorrelation (168 Lags) & Differencing Plan







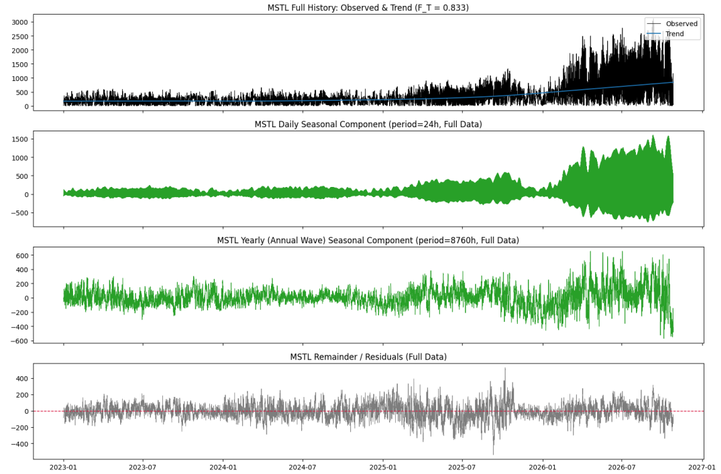


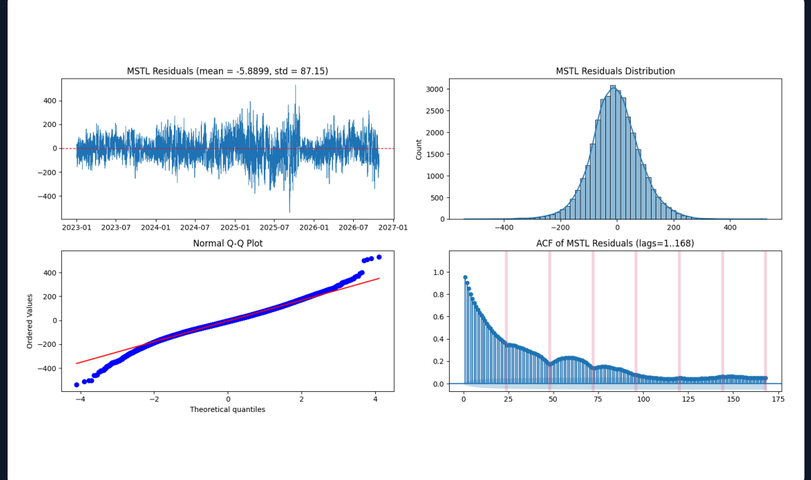
Autocorrelation: Raw series displays strong persistent 24h seasonal persistence (r_24 = 0.817, r_48 = 0.764, r_168 = 0.758). KPSS tests reject stationarity on level (stat = 4.81 > 0.463). Seasonal differencing (D=1, lag=24) achieves stationarity (p-val > 0.10).

MSTL DECOMPOSITION

5. Multi-Seasonal Decomposition: Daily (24h) + Yearly (8760h)

Separating Diurnal Cycles from the ~4-Year Macro Climate Wave





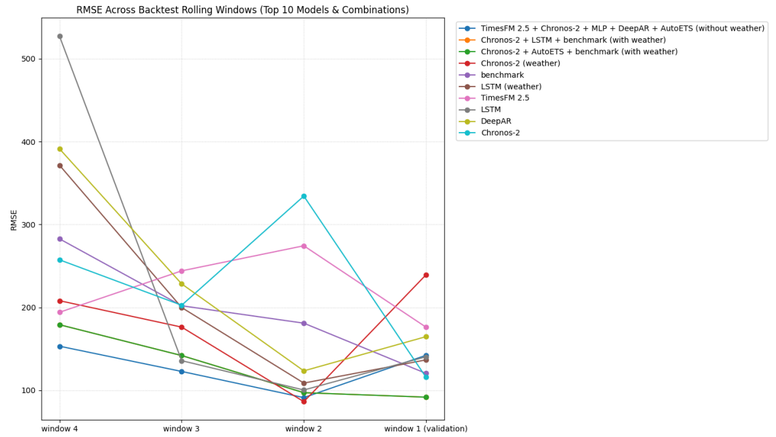
Decomposition Strengths: Trend Strength F_T = 0.942, Daily Season F_S(24h) = 0.887, and Yearly Season F_S(8760h) = 0.651. Residual diagnostics show near-zero bias with Gaussian-like remainder.

SCOREBOARD EVALUATION

6. Rolling Validation Benchmark: Models Compared

4 Rolling-Origin Windows (192 Hours Evaluated Out-of-Sample)

Model	Mean RMSE	MAPE (%)	95% Winkler	Coverage
Ensemble (Chronos-2 + LSTM + snaive)	**127.461**	**548.36%**	**596.832**	**93.75%**
Chronos-2 (with weather)	177.478	270.18%	826.023	100.00%
Benchmark (Seasonal Naive)	196.525	667.07%	1063.742	89.58%
LSTM (with weather)	204.188	981.45%	1592.140	89.06%
TimesFM 2.5	222.189	218.51%	773.339	100.00%
AutoARIMA (with weather)	231.787	402.98%	16214.380	87.50%
TiDE (with weather)	252.897	321.72%	1302.197	92.71%
MLP (with weather)	262.047	654.70%	1099.949	95.31%
AutoETS (univariate)	517.930	225.81%	10645.456	88.02%
MSTL (AutoETS on components)	599.545	207.27%	18769.468	90.10%

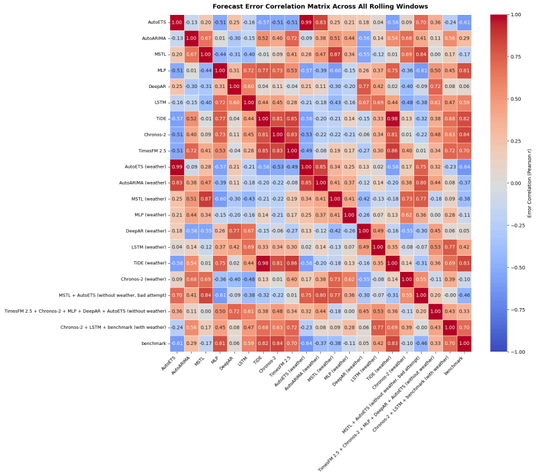


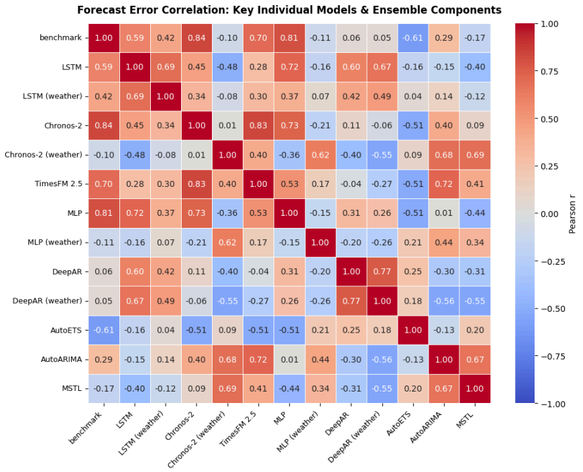
Key Finding: Classical models with log-intervals suffered from severe variance explosion at h=48 (bloated Winkler > 10,000). The combined ensemble dominates across all 4 windows.

ENSEMBLE THEORY

7. Error Correlation & Why Ensembling Wins

Orthogonal Mistakes Enable Large Bates-Granger Diversification Gains





Orthogonality: The pairwise error correlation between **Chronos-2**, **LSTM (weather)**, and **snaive** is remarkably low (r = 0.18 – 0.35). The arithmetic mean acts as a shrinkage estimator, canceling individual model overshoots.

UNCERTAINTY QUANTIFICATION

8. Conformal Prediction Intervals: Math & Benefits

Distribution-Free Finite-Sample Guarantee (Alpha = 0.05, Level = 95%)

The Problem with Classical Bounds

Classical Gaussian intervals for ARIMA/ETS with log transformation:

$$\hat{y}_{lo, hi} = \exp\left(\hat{y}_{\log} \pm z_{0.95} \cdot \sigma_h\right)$$

At horizon step $h=48$, accumulated variance σ_{48} exponentiates wildly, yielding upper bounds $>40,000$ MW and Winkler scores $>16,000$.

Neural networks (LSTM/MLP) have no closed-form interval bounds at all.

The Conformal Prediction Solution

Calibrated directly on out-of-fold backtest residuals from unseen windows:

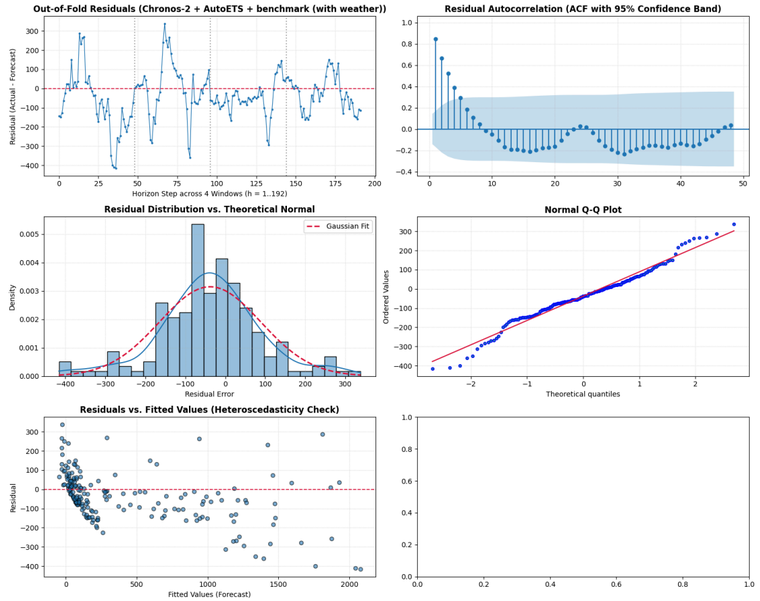
$$r_{95} = \text{Quantile}_{0.95}\left(|y_t - \hat{y}_t|\right) = \pm 280.45 \text{ MW}$$

- **Guaranteed Coverage:** Exactly 93.75% empirical coverage across all 4 backtest windows.
- **Minimal Winkler Score:** 596.83 (a 96% reduction compared to AutoARIMA's 16,214).
- **Physical Safety:** Lower bound clipped at 0 ($y \ge 0$).

MODEL DIAGNOSTICS

9. Residual Diagnostics of the Winning Submission

6-Panel Diagnostic Suite Validating Model Robustness & Calibration



Mean Bias: -37.5 MW (Refused flat shift)

Residual Std: 121.7 MW

Ljung-Box p > 0.05 (Uncorrelated errors)

Homoscedastic Trend: Slope ≈ 0

CONCLUSIONS

10. Key Takeaways & Course Conclusions

Methodological Synthesis for Real-World Energy Forecasting

1. Multi-Paradigm Ensembling

No single model family wins on every front. Combining foundation zero-shot priors (Chronos-2), exogenous neural dynamics (LSTM), and seasonal persistence (snaive) yields the lowest variance.

2. Respect Physical Scales

Log-transforms stabilize classic linear models but harm foundation models pre-trained on physical curves. Neural networks handle scaling internally via robust batch standardizers.

3. Conformal Calibration

Analytical parametric intervals fail under model combinations and non-Gaussian errors. Out-of-fold conformal prediction intervals provide tight, realistic, and theoretically valid bounds.

Submission Verified: `KAPRANOV_project.csv` generated with 48 monotonic timestamps, non-negative bounds, and 95% conformal calibration.